MUHAMMAD BILAL TARIQ

Computer Science Student — NUST Islamabad

Islamabad, Pakistan | iambilaltariq2005@gmail.com | GitHub | LinkedIn | Portfolio

PROFILE

Computer Science student at NUST Islamabad with hands-on experience building and shipping full-stack and AI-integrated applications, from database-backed systems to LLM-powered products. Proficient in Python, C++, Java, React, Node.js/Express, and PyTorch/TensorFlow. Focused on writing code that ships, measuring outcomes, and applying ML to practical problems. Full project portfolio, write-ups, and source code available via the GitHub and Portfolio links above.

EDUCATION

BS Computer Science, NUST Islamabad (H-12) 2024 — Present

- Relevant coursework: Data Structures & Algorithms, OOP, Operating Systems, Database Systems, Theory of Automata, Web Engineering
- Engaged in departmental projects and collaborative software development

TECHNICAL SKILLS

- Languages: Python, C++, Java, JavaScript
- Web & Frameworks: HTML/CSS/JS, React, Express, Node.js, REST APIs
- AI/ML: PyTorch, TensorFlow, scikit-learn, LLM/prompt engineering, OpenCV
- Databases: MySQL, MongoDB
- CS Core: Data Structures, Algorithms, OOP, Operating Systems
- Tools: Git/GitHub, Vercel, Render, VS Code, Postman

PROJECTS

PromptForge : AI-Powered Prompt Optimization Web App 2026

React, Vite, Tailwind CSS, Node.js/Express, MongoDB Atlas, Groq API (Llama 3.3 70B), Google OAuth

- Designing and building a full-stack SaaS tool that rewrites and scores user prompts for clarity and effectiveness, using Groq's Llama 3.3 70B for real-time optimization
- Architected the system end-to-end: React/Vite/Tailwind frontend on Vercel, Express API on Render, MongoDB Atlas for persistence, and Passport.js for Google OAuth

NovaBanque : Full-Stack Banking Web Application 2026

Node.js, Express, MySQL, Vanilla JS — Team project (3 members), CS 236 Web Engineering

- Built a full banking platform supporting accounts, transfers, and statements as the CS 236 Web Engineering final project, delivered with a full technical report and a 17-slide presentation
- Improved code quality and UX by removing all debug console statements and replacing native alert() popups with a custom toast notification system across the codebase

Library Management System : Catalog & Fine-Tracking Database System 2025

Python, MySQL, OOP Principles, Normalized (3NF) Schema Design

- Built a modular, OOP-based catalog system handling full CRUD operations for book inventory, borrower logs, and user cards
- Designed dynamic algorithms to calculate late-return fines and update member borrowing states on login
- Normalized the MySQL schema up to 3NF to eliminate record redundancy and ensure data integrity across transactions

Qalam Clone : Full-Stack Web Application 2024

React, Express, Node.js, MongoDB, REST API

- Built a feature-rich clone of NUST's Qalam academic platform, replicating authentication, course enrollment, grade tracking, and assignment submission
- Designed a RESTful API backend with Node.js/Express connected to a MongoDB database

Smart Fashion Store : AI-Powered Fashion E-Commerce Platform 2025

Python, Flask, SQLAlchemy, SQLite, HTML/CSS

- Built a Flask-based e-commerce platform with user authentication, product browsing, and an admin dashboard for managing products and recommendations
- Implemented an AI-powered outfit recommendation engine, modeling users, products, and recommendations with SQLAlchemy

More projects, live demos, and write-ups: bilal-portfolio-swart.vercel.app

INTERESTS

- AI/ML & LLMs: supervised/unsupervised learning, transformer architectures, prompt engineering, RAG
- Computer Vision: image classification, object detection, CNNs, OpenCV and PyTorch-based vision models
- Computer Security: secure software design, cryptography fundamentals, and the intersection of security with AI/ML systems